Junuk Cha

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Interests

Computer vision based on deep learning related to 3D humans (e.g., 3D avatar reconstruction, 3D scene reconstruction, 3D human motion generation, and interacting 3D human–object motion generation), robot pose estimation, humanoid motion generation, text-guided 2D image editing, and audio-driven talking head synthesis.

Education

Ulsan National Institute of Science and Technology *Ulsan, South Korea*
Combined MS/PhD. at Artificial Intelligence Graduate School *Mar. 2021 - Feb. 2025*
Supervisor: Professor Seungryul Baek

Ulsan National Institute of Science and Technology *Ulsan, South Korea*
Bachelor at Department of Mechanical Engineering *Mar. 2015 - Feb. 2021*
Sergeant, Republic of Korea Army, Honorably Discharged (Jan. 2018 – Sep. 2019)

Work Experience

Senior Researcher at NC AI (Digital Twin R&D), Pangyo, Republic of Korea *Nov. 2026 - Present*
Digital Twin Research

Postdoc at KAIST, Daejeon, Republic of Korea *Aug. 2025 - Aug. 2026*
Egocentric Humanoid Control (Perception & Planning), Multiple human-object interaction, Robot Pose Estimation, Non-line-of-sight *InnoCORE Research Group for Bio-Embodied Physical AI, led by DGIST*
PI: Tae-Hyun Oh

Postdoc at KETI, Pangyo, Republic of Korea *Mar. 2025 - Jul. 2025*
OpenFS: Multi-Hand-Capable Fingerspelling Recognition with Implicit Signing-Hand Detection and Frame-Wise Letter-Conditioned Synthesis (CVPR 2026)
PI: Han-Mu Park

Research intern at Inria, Nice, France *Sep. 2024 - Nov. 2024*
EmoTalkingGaussian: Continuous Emotion-conditioned Talking Head Synthesis (arXiv 2025)
Collaborators: Seongro Yoon, Valeriya Strizhkova, Francois Bremond

Research Scientist/Engineer Intern at Adobe, San Jose, CA, USA *Feb. 2024 - May 2024*
Text2Relight: Creative Portrait Relighting with Text Guidance (AAAI 2025)
Collaborators: Jae Shin Yoon, Mengwei Ren, He Zhang, Krishna Kumar Singh, Yannick Hold-Geoffroy, David Seunghyun Yoon, HyunJoon Jung

Skills

Programming Python (PyTorch, NumPy, Pandas, etc.), C/C++.
Miscellaneous Isaac Gym, Isaac Sim, Linux, Shell (Bash/Zsh), LaTeX (Overleaf/R Markdown), Git, Docker.
Soft Skills Time Management, Teamwork, Problem-solving, Documentation, Self-Motivation, Eager Learner.

Publications (International)

* Equal contribution, † Corresponding author or co-last author

- **RoboFS: Scalable Data Generation and Text-to-Action Learning for Robot Fingerspelling**
  Jihyeon Kim, **Junuk Cha**†.
  in Proc. of IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV)** workshop, Orlando, USA, 2027.
- **DRIP: Self-Improving Robot State Estimation In-the-Wild via Dense Robot-Fitting In the LooP**
  Ohn Kim, Youwang Kim, **Junuk Cha**, Tae-Hyun Oh†.
  IEEE Robotics and Automation Letters **(RA-L)**, 2026.
- **OpenFS: Multi-Hand-Capable Fingerspelling Recognition with Implicit Signing-Hand Detection and Frame-Wise Letter-Conditioned Synthesis**
  Junuk Cha, Jihyeon Kim, Han-Mu Park†.
  in Proc. of IEEE Conf. on Computer Vision and Pattern Recognition **(CVPR)**, Denver, USA, 2026.

- **CoT-Pose: Chain-of-Thought Reasoning for 3D Pose Generation from Abstract Prompts**
  Junuk Cha*, Jihyeon Kim*.
  in Proc. of IEEE/CVF International Conference on Computer Vision **(ICCV)** workshop, Honolulu, USA, 2025.
- **BIGS: Bimanual Category-agnostic Interaction Reconstruction from Monocular Videos via 3D Gaussian Splatting**
  Jeongwan On, Kyeonghwan Gwak, Gunyoung Kang, **Junuk Cha**, Soohyun Hwang, Hyein Hwang, Seungryul Baek†.
  in Proc. of IEEE Conf. on Computer Vision and Pattern Recognition **(CVPR)**, Nashville, USA, 2025.
- **EmoTalkingGaussian: Continuous Emotion-conditioned Talking Head Synthesis**
  Junuk Cha, Seongro Yoon, Valeriya Strizhkova, Francois Bremond, Seungryul Baek†.
  arXiv 2025.
- **Text2Relight: Creative Portrait Relighting with Text Guidance**
  Junuk Cha, Mengwei Ren, Krishna Kumar Singh, He Zhang, Yannick Hold-Geoffroy, Seunghyun Yoon, HyunJoon Jung, Jae Shin Yoon†, Seungryul Baek†.
  in Proc. of Association for the Advancement of Artificial Intelligence (**AAAI**), Philadelphia, USA, 2025.
- **3DGS-based Bimanual Category-agnostic Interaction Reconstruction**
  Jeongwan On, Kyeonghwan Gwak, Gunyoung Kang, Hyein Hwang, Soohyun Hwang, **Junuk Cha**, Jaewook Han, Seungryul Baek†.
  in Technical Report of European Conference on Computer Vision (**ECCV**), Milan, Italy, 2024.
- **Text2HOI: Text-guided 3D Motion Generation for Hand-Object Interaction**
  Junuk Cha, Jihyeon Kim, Jae Shin Yoon†, Seungryul Baek†.
  in Proc. of IEEE Conf. on Computer Vision and Pattern Recognition **(CVPR)**, Seattle, USA, 2024.
- **VLM-PL: Advanced Pseudo Labeling Approach for Class Incremental Object Detection via Vision-Language Model**
  Junsu Kim, Yunhoe Ku*, Jihyeon Kim*, **Junuk Cha**, Seungryul Baek†.
  in Proc. of IEEE Conf. on Computer Vision and Pattern Recognition **(CVPR)** workshop, Seattle, USA, 2024.
- **3D Reconstruction of Interacting Multi-Person in Clothing from a Single Image**
  Junuk Cha, Hansol Lee, Jaewon Kim, Bao Truong, Jae Shin Yoon†, Seungryul Baek†.
  IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV)**, Waikoloa, USA, 2024.
- **Dynamic Appearance Modeling of Clothed 3D Human Avatars using a Single Camera**
  Hansol Lee, **Junuk Cha**, Yunhoe Ku, Jae Shin Yoon†, Seungryul Baek†.
  arXiv 2023.
- **HOReeNet: 3D-aware Hand-Object Grasping Reenactment**
  Changhwa Lee, **Junuk Cha**, Hansol Lee, Seongyeong Lee, Donguk Kim, Seungryul Baek†.
  arXiv 2022.
- **Multi-Person 3D Pose and Shape Estimation via Inverse Kinematics and Refinement**
  Junuk Cha, Muhammad Saqlain, GeonU Kim, Mingyu Shin, Seungryul Baek†.
  in Proc. of European Conference on Computer Vision **(ECCV)**, Tel Aviv, Israel, 2022.
- **Learning 3D Skeletal Representation From Transformer for Action Recognition**
  Junuk Cha, Muhammad Saqlain, Donguk Kim, Seungeun Lee, Seongyeong Lee, Seungryul Baek†.
  in IEEE Access 2022 (IF: 3.367).
- **3DMesh-GAR: 3D human body mesh-based method for group activity recognition**
  Muhammad Saqlain, Donguk Kim, **Junuk Cha**, Changhwa Lee, Seongyeong Lee, Seungryul Baek†.
  in Sensors 2022 (IF: 3.576).
- **Towards Single 2D Image-level Self-supervision for 3D Human Pose and Shape Estimation**
  Junuk Cha, Muhammad Saqlain, Changhwa Lee, Seongyeong Lee, Seungeun Lee, Donguk Kim, Won-Hee Park, Seungryul Baek†.
  in Applied Sciences 2021 (IF: 2.679).

Patents

- **Video-based 3D mesh restoration method and system therefor, 2024.**
  영상 기반의 3차원 메쉬 복원 방법 및 이를 위한 시스템
  Applicant: CJ Olive Networks and UNIST
  Korean Patent Application No. 1020240169695, Filed Nov. 25, 2024
  DOI: 10.8080/1020240169695
- **Method for restoring interaction of multiple objects expressed in a 2D image to 3D, 2024.**
  2차원 이미지에 표현된 복수 객체의 상호작용을 3차원으로 복원하는 방법
  Applicant: UNIST

Korean Patent No. 1029495260000, Registered Apr. 2, 2026
DOI: 10.8080/1020240050701

- **Artificial intelligence based worker detection device and method using thermal imaging, 2023.**
  열화상 영상을 이용한 인공지능 기반 작업자 검출 장치 및 방법
  Applicant: Hyundai and UNIST
  Korean Patent Application No. 1020230100243, Filed Aug. 1, 2023
  DOI: 10.8080/1020230100243

Projects

Development of vision recognition module for ARTG control *Ministry of Oceans and Fisheries*
UNIST, Ulsan, South Korea *Jun. 2023 - Dec. 2023*

- Detecting object and predicting multi agent's future trajectory with velocity.

Development of an optimized human detection system algorithm for production factory environments *Hyundai*
UNIST, Ulsan, South Korea *Jun. 2022 - Dec. 2023*

- Detecting human segmentation masks and bounding boxes.
- Improving performance for corner cases such as occluded people.

Awards

- **Industry-Academia Best Paper Award, LG Electronics**, 2026 Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE), Jeju, Korea, 2026.
  MotionGlue: A Framework for Natural Transition Generation Between Heterogeneous Human Motions
  산업체 관심논문상(LG전자), 2026년도 대한전자공학회 하계종합학술대회, 제주, 2026.
- The 8th Workshop on Observing and Understanding Hands, **ECCV 2024, 1st place in HANDS workshop challenge**
- The 7th Workshop on Observing and Understanding Hands, **ICCV 2023, 3rd place in HANDS workshop challenge**

Academic Activities

- **Outstanding Reviewer at CVPR 2026**
- **Program Committee (Reviewer) at CVPR 2025, 2026**
- **Program Committee (Reviewer) at ICCV 2025**
- **Program Committee (Reviewer) at ECCV 2026**
- **Program Committee (Reviewer) at ICLR 2025, 2026**
- **Program Committee (Reviewer) at AAAI 2023, 2024, 2025, 2026, 2027**
- **Program Committee (Reviewer) at NeurIPS 2024, 2026**
- **Program Committee (Reviewer) at BMVC 2023, 2024, 2025, 2026**
- **Program Committee (Reviewer) at CoRL 2026**

Talks

- **Multi-Person 3D Pose and Shape Estimation via Inverse Kinematics and Refinement**, CV/ML Session, 2022 Fall Conference of the Korean Artificial Intelligence Association (KAIA), Nov. 18, 2022.

Teaching Assistant & Mentoring

- **Mentoring**
  - 2nd Ulju AI 4.0 Studio at UNIST, Ulsan, South Korea, Mar. 2022 - Nov. 2022.
  - Mentored an undergraduate student on a non-line-of-sight perception task at KAIST, Daejeon, South Korea, Dec. 2025 - Aug. 2026.

- **TA**
  - Navy AI professionals training process TA, Jinhae, South Korea, 12 Sep. 2023.
  - 5th AI Novatus Academia PBL (Computer Vision part) at UNIST AI Innovation Park, Ulsan, South Korea, Mar. 2023 - Jun. 2023.
  - 1st Gyungnam AI Novatus Academia (Computer Vision part), Masan, South Korea, 15 Apr. 2022.

- 3rd AI Novatus Academia (Computer Vision part) at UNIST AI Innovation Park, Ulsan, South Korea, 25 Feb. 2022.

- **Lecture TA**
  - 2023 Fall AI51601 Computer Vision.
  - 2021 Fall ITP11701 Introduction to AI Programming 2.